The plane containing the vibrations of the electric field vector is known as the ***plane of vibration*** *ABCD* as shown in Figure 7.26. The plane perpendicular to the plane of vibration is known as the ***plane of polarisation*** *EFGH*. Both the plane of vibration and the plane of polarisation contain the direction of propagation of light.

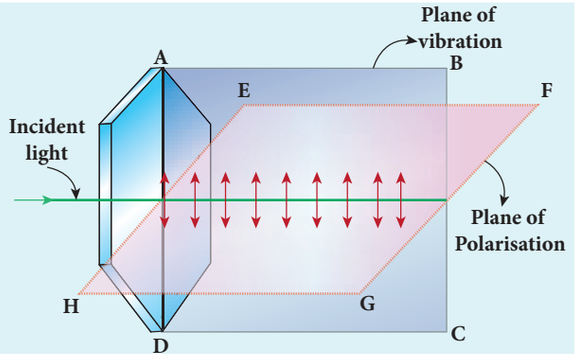


Figure 7.26 Plane of vibration and plane of polarisation

The Table 7.3 consolidates few characteristics of polarised and unpolarised light.

7.5.2 Polarisation Techniques

The polarised light can be obtained from unpolarised light by several techniques. Here, we are discussing the four methods. They are,

(i) polarisation by selective absorption
(ii) polarisation by reflection
(iii) polarisation by double refraction
(iv) polarisation by scattering.

7.5.3 Polarisation by selective absorption

Selective absorption is the property of a material which transmits waves whose electric field vibrations are in a plane parallel to a certain direction of orientation and absorbs all other other vibrations. The *polaroids* (or) *polarisers* are thin commercial sheets which make use of the property of selective absorption to produce plane polarised light. Selective absorption is also called as *dichroism*.

In 1932, an American scientist Edwin Land developed polarisers in the form of sheets. Tourmaline is a natural polarising material. Polaroids are also made artificially. It was discovered that small needle shaped crystals of quinine iodosulphate have the property of polarising light. A number of these crystals with their axes parallel to one another packed in between two transparent plastic sheets serve as a good polaroid. Recently, new types of polaroids are prepared in which thin film of polyvinyl alcohol is used. These are colourless crystals which transmit more light, and give better polarisation.

Table 7.3 Characteristics of polarised light and unpolarised light

S.No	Polarised light	Unpolarised light
1	Consists of waves having their electric and magnetic field vibrations in a single plane normal to the direction of ray.	Consists of waves having their electric and magnetic field vibrations in all directions normal to the direction of ray.
2	Asymmetrical about the ray direction.	Symmetrical about the ray direction.
3	It is obtained by converting unpolarised light using polaroids.	Produced by conventional light sources.

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